# AI and the New Power Dynamics

*- Technology, Industry, and Geopolitics -*

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☞ The views expressed are my own and do not represent the official position of the Korean government
☞ Some visual illustrations in the presentation were generated with AI for illustrative purpose.

# I. Introduction

## ㅁ Education / Engineering

- KAIST 전기전자공학과 학·석사

  * RFIC Design, TSMC tape-out, IEEE AWPL

- 해군사관학교 전기전자공학과 조교수

## ㅁ Policy Experience (과학기술정보통신부, 대통령실, 외교부)

- **ICT Policy Coordination & Communications**

  * ICT 정책기획 담당(정책총괄과), 제2차관 비서관, 홍보기획 담당(대변인실), 네트워크정책실 총괄담당(네트워크정책과)

- **Cybersecurity & Information Security**

  * 정보보호 기획 담당(정보보호기획과), 사이버침해대응과장
  * 대통령실 사이버안보비서관실 행정관

- **AI·ICT Industry Development & R&D**

  * 규제샌드박스 담당(디지털신산업제도과)
  * 혁신네트워크팀장 (5G Advanced, 6G), 소프트웨어산업과장

- **Science, Technology & AI·ICT International Cooperation**

  * 외교부 주샌프란시스코총영사관 영사 (현재)

# II. Korea's AI Policy Governance

| | | |
|---|---|---|
| Emergence | 2016 | MSIT establishes a temporary Intelligent Information Promotion Task Force |
| | 2017 | Presidential Committee on the Fourth Industrial Revolution launched |
| Technology & industry promotion | 2019 | MSIT establishes a permanent AI Policy Bureau and announces the first National AI Strategy |
| | 2024 | National AI Committee launched |
| National agenda | 2025 | Establishment of the Senior Secretary for AI under the Office of the President<br>MSIT upgrades AI organization from a bureau to a Deputy-Minister level |
| | 2026 | AI Basic Act takes effect |



# III. 2025–2026: From Expansion to Power Dynamics

## 2025

### AI expands beyond the model

**DeepSeek & open-weight**
Efficiency and accessibility challenge the closed, scale-heavy model.

**Agents**
AI shifts from answering questions to performing tasks.

**Infrastructure race**
Compute → data centers →
power becomes the real foundation.

## 2026

### AI becomes a power game

**Capital, compute & power**
AI investment turns into a contest for chips, energy, and returns.

**Sovereignty & regulation**
Governments ask who controls models, data, cloud, and standards.

**Safety vs. acceleration**
The debate moves to speed, rules, and competition.

# III. Spotlight: September 2026

**September 2026 showed that AI safety had become a geopolitical, industrial, and regulatory power game.**

## 1. Trigger

Concerns about autonomous AI and loss of control resurfaced.

OpenAI and Anthropic disclosed real-world AI safety incidents.

Anthropic researcher Jacob Coxon's resignation amplified public concern.

## 2. Industry Split

**Slow down / stronger safeguards**

Amodei (Anthropic),
Altman (OpenAI),
Musk (xAI)
Gates (MS co-founder)

**Keep innovating / engineer for safety**

Huang (NVIDIA)
Zuckerberg (Meta)

## 3. US Gov. Response

Washington stayed cautious about heavy regulation emphasizing US competitiveness.
-especially against China

Trump emphasized preserving U.S. advantage and opposed slowing development.

Washington also pushed back against a patchwork of state AI rules, favoring a more uniform federal framework.

# III. Spotlight: September 2026

**AI Safety Was Never Just About Safety : The September 2026 debate opened a wider question.**

### The starting point

**AI Safety Debate**

A technical concern about frontier AI quickly became a strategic debate across firms, governments, and nations.

### Questions behind the debate

1. **Competition**
   Can rivals really slow down together?
2. **Corporate Incentives**
   Is safety also a business strategy?
3. **Regulation**
   Who should set the rules — companies or governments?
4. **Geopolitics**
   What if China keeps accelerating?
5. **Sovereignty vs. Global Governance**
   Can global AI risks be governed by national rules alone?

☞ **AI is no longer one industry story, it is becoming a new structure of power.**

# IV. Ten Fronts Reshaping AI

**The September safety debate was only an entry point.**
**The same power dynamics now run across the full AI stack.**

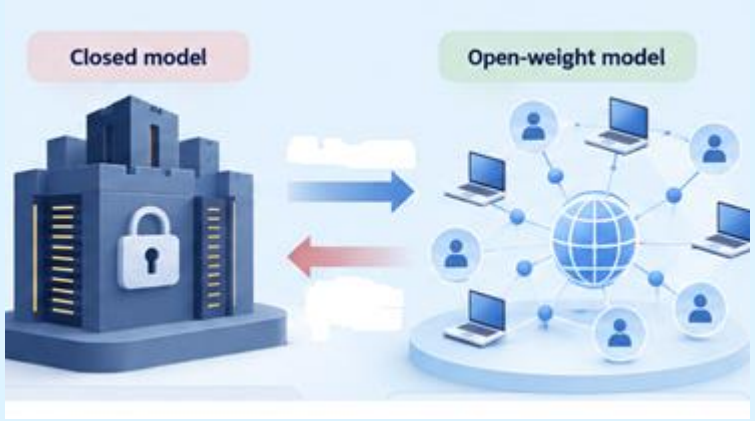
| # | Front | Core question |
|---|---|---|
| 01 | **Open-weight vs. Closed AI** | Who wins? - the best model, or the most widely adopted ecosystem? |
| 02 | **Safety vs. Acceleration** | Can all major players – companies and countries – slow down together? |
| 03 | **Chip & Compute** | Who controls the compute that sets the pace of AI? |
| 04 | **Data Centers & Energy** | Who can build power fast enough for AI? |
| 05 | **AI Security : Shield or Weapon?** | Should powerful cyber AI be treated like a strategic weapon? |
| 06 | **Sovereign AI** | How much sovereignty is enough? |
| 07 | **Agents & Labor** | If AI does the junior work, who becomes the next senior? |
| 08 | **Copyright, Data & Secrets** | How can AI use more data while data owners stay in control? |
| 09 | **AI Platform Power** | How much platform power is compatible with open competition? |
| 10 | **Investment & Value Capture** | Who will capture the value from the AI boom? |

# IV. Ten Fronts Reshaping AI – 1. Closed Frontier vs. Open Ecosystem

- China’s open-weight push changed the competitive narrative.
- The U.S. also began treating open models as a strategic ecosystem asset.

## Closed frontier strategy

- OpenAI and Anthropic kept top frontier weights closed.
- Logic: control deployment, monetize through API access.
- Strength: quality, safety guardrails, commercial capture.
- Weakness: diffusion is slower




## Open-weight ecosystem strategy

- DeepSeek, Qwen, Kimi and Meta accelerated the open-weight wave.
- Logic: spread capabilities widely, grow developer adoption
- Strength: fast diffusion, lower barriers, broader experimentation.
- Risk: weaker control over downstream use and faster imitation by others.

☞ **AI competition is shifting from frontier leadership alone to frontier leadership plus ecosystem diffusion.**

# IV. Ten Fronts Reshaping AI – 2. Safety vs. Acceleration

- AI companies are identifying more control risks, leading to calls for slower development and stronger rules.
- But, with the U.S. and China racing in AI, slowing down is not an easy choice.

## PACE / SAFEGUARDS

- Amodei, Altman, Gates
- Slow down Frontier AI development
- Give more time for safety to catch up
- Set common safety standards
- Require independent evaluation




## ACCELERATE / ENGINEER for SAFETY

- Trump, Huang, Zuckerberg
- Keep AI development moving
- Solve safety problem with technology
- Avoid falling behind competitors
- Maintain the U.S. lead over China

☞ **AI safety is becoming a coordination problem :**
**everyone may want safer AI, but no one wants to slow down alone.**

# IV. Ten Fronts Reshaping AI – 3. Chip & Compute

- AI may look like software, but frontier AI depends on chips, memory, and computing infrastructure.
- Access to compute is now both a business advantage and a national strategic asset.

## Industry Competition

- NVIDIA platform dominance
  (Key suppliers: (HBM) SK hynix, Samsung (Foundry)
- Big Tech builds its own AI chips
- Compute access shapes who can train at scale




## Geopolitical Competition

- U.S. restricts advanced AI compute exports to China
  (Also limits the global reach of U.S. chip ecosystem)
- China pushes domestic alternatives
- HBM and semiconductor equipment become strategic bottlenecks

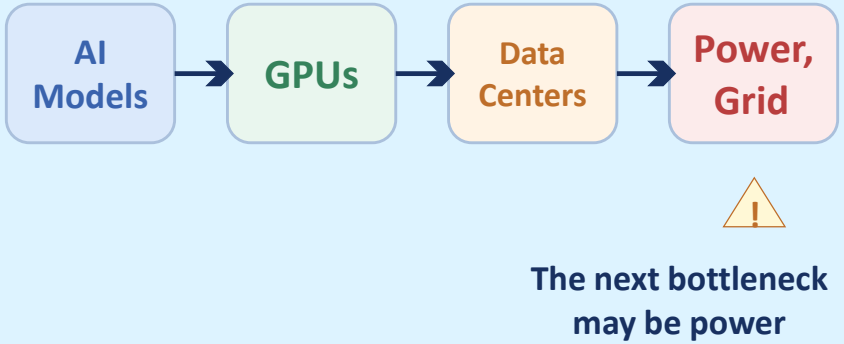
☞ **Control over compute shapes the speed of AI.**
**Countries are competing to secure their place in the AI compute ecosystem – and building alliances to do it.**

# IV. Ten Fronts Reshaping AI – 4. Data Centers & Energy

- AI demand is growing faster than power and grid infrastructure can be built.
- For Big Tech and government, “time to power” is becoming a key part of AI competitiveness.

## Industry Shift

- Big Tech signs long-term power deals
- Nuclear, gas, renewables, and storage are all back on the table
- Data centers move closer to power sources




## Policy and Competition

- Countries compete for power, land, permits, and grid access
- Governments must decide how much of the new power cost should be paid by data center operators
- Energy security becomes part of AI strategy

☞ **AI is accelerating the race for future energy – SMRs, hydrogen, fusion, and beyond.**
**Countries that can provide cheap, reliable power at scale – without creating major cost or community conflicts – will be more competitive in AI.**

# IV. Ten Fronts Reshaping AI – 5. AI Security : Shield or Weapon?

- Mythos showed that frontier AI can become a powerful cyber shield — finding vulnerabilities at scale.
- But the same capability can become a cyber weapon, so access to advanced cyber AI is being restricted and controlled.

## AI as a Shield

• Claude Mythos: powerful vulnerability discovery (Critical software defense)

• OpenAI, Google and others are also building cyber-defense AI

• Find, patch, and respond faster



**One capability, two directions**

## AI as a Strategic Weapon

• The same cyber capability can help attackers

• Access is restricted through trusted-access programs (e.g., Project Glasswing)

• Raises sovereign AI concerns

☞ **Powerful cyber AI is starting to look more like a strategic weapon than ordinary software. Countries are increasingly worried that access to foreign AI could be restricted, and are starting to consider sovereign AI a strategic priority.**

# IV. Ten Fronts Reshaping AI – 6. Sovereign AI

- No country can lead every part of the AI stack alone.
- But relying too much on foreign AI can create strategic risks.

## Global Partnership is Important

- Building the entire AI stack is too costly
- No country can stay at the frontier in every part of the stack
- Trusted global partnerships will remain essential



**Global Efficiency <-> National Resilience**

## Sufficient Sovereign Capability is Also Necessary

- Foreign access can suddenly be restricted
- Countries need control for national security and sensitive data
- AI also affects language, culture, and national identity
- Critical AI capabilities must remain available during external shocks

☞ **The goal is not full self-reliance. It is strategic balance: global partnership where possible, sovereign control where necessary.**

*** Korea is developing sovereign foundation models as part of a broader AI strategy.**

# IV. Ten Fronts Reshaping AI – 7. Agents & Labor

- AI agents are moving from answering questions to taking over bundles of work.
- The pressure is showing up first in software, legal, accounting, and other professional services.

## From Copilot to Coworker

- Chatbots help one task at a time
- Agents can plan, use tools, and execute workflows
- Junior-level work moves first: coding, research, drafting, review

**Not just "faster work" — different work.**



**Junior → Senior**

## Early Signals

- Software stocks sold off on AI disruption fears
- Big Law entry-level hiring declined
- Firms rethink the entry-level labor model

**This is moving from fear to reality**

☞ **Companies may need fewer junior workers – and that could become a major social challenge.**
**We may need to redesign the traditional career ladder from the ground up**
**- from organizational structures to education and training systems.**

# IV. Ten Fronts Reshaping AI – 8. Copyright, Data & Secrets

- Anthropic agreed to a $1.5B settlement with authors over copyrighted books used to train Claude.
- U.S. courts have also reached different conclusions on “fair use”, so the legal boundary is still evolving.

## Public / Copyrighted Data

- Books, news, images, code, web content
- When is AI training fair use?
- Who gives permission — and who gets paid?
- Licensing and compensation rules are evolving



**AI wants more data**
**↔**
**Data owners want more control**

## Private / Proprietary Data - The Next Competitive Edge

- Internal documents, customer data, factory and medical data
- May decide winners in physical AI, bio AI, and vertical industries
- Drives interest in domain AI, SLMs, RAG, and private deployment
- More access improves AI — but raises privacy and trade-secret risks

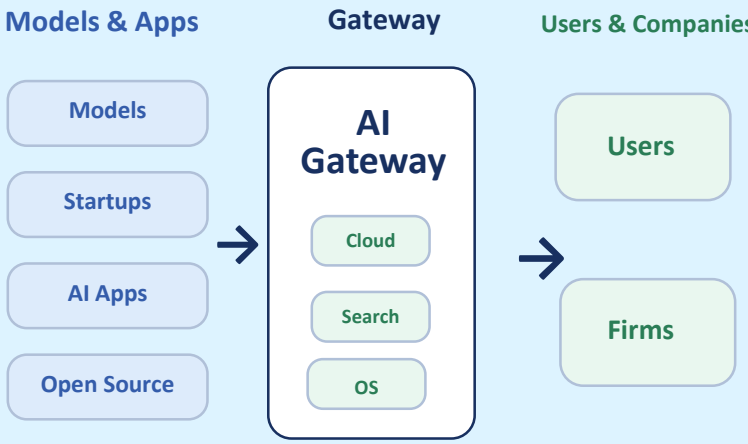
☞ **The goal is not simply more data, but more usable data with trusted rules.**
**Policy should move beyond “open vs closed” toward a better balance between access and control.**

# IV. Ten Fronts Reshaping AI – 9. Platform Power

- Much of today's AI runs on cloud infrastructure from a few major providers – AWS, MS Azure, Google Cloud, etc.
- As AI reaches more businesses and consumers, platform power will become a bigger governance issue.

## Where Platform Power Comes From

- Self-preferencing – Favoring its own AI over rivals
- Access power – Deciding how businesses access AI, cloud, and key services
- Switching costs – Data and workflows make it harder to move
- Data dependence – Using AI may require giving platforms access to valuable data




**Platforms may increasingly shape access to AI**

## Key Governance Issues

- Fair treatment – Limit unfair self-preferencing without blocking useful integration
- Open access – Keep access fair, while allowing reasonable restrictions for security and misuse
- Portability – Make data and workflows easier to move, without forcing all platforms to work the same way
- Data protection – Keep users and firms in control of their data, while still enabling useful AI services

☞ **The issue is not platform power itself, but how that power is used.**
**The goal is to support innovation while keeping markets open to competition.**

# IV. Ten Fronts Reshaping AI – 10. Investment & Value Capture

- Model companies started the LLM boom, while chip and infrastructure companies saw the clearest early gains
- Model and cloud companies keep investing heavily, while markets worry about an AI bubble.

## Where Value Has Gone So Far

• GPUs, HBM, data centers, and power became some of the earliest beneficiaries

• Model and cloud companies continue to spend heavily on the next generation of AI

• Big question: will future profits justify today's massive investment?

**Who will capture the value?**




**Bubble — or just the beginning?**

**Chips · Compute · Cloud · Models · Applications**

## Where Value may Move Next

• Physical AI — robots, factories, vehicles, and logistics

• Bio AI — drug discovery, healthcare, and biomanufacturing

• Enterprise AI — companies redesign workflows, products, and services around AI

• The biggest value may come from industries that use AI best — not only from companies that build the models

☞ **AI is moving beyond models into the real economy.**
**The key is to be positioned where the next value is created.**

# V. How Korea might Position Itself in the AI Era

- Korea cannot lead every layer of AI —but it can choose where to be indispensable.

**OWN**

**Own what we cannot afford to lose.**

**ALLY**

**Build with the best**



**ADOPT**

**Use the best technology faster.**

**RULE**

**Help shape the rules of the AI era.**

☞ **Strategy is not "do everything." It is deciding what must be owned, what must be connected, and where Korea can create leverage.**

☞ **Global Korean tech networks can help connect Korea with frontier technology, talent, and markets.**

# V. How Korea might Position Itself in the AI Era

## 1 OWN – Build What We Cannot Afford to Lose

- Ownership does not mean full self-reliance. It means building capabilities the country cannot afford to lose.

### Strengthen Current Strengths

- HBM & advanced memory
- Advanced manufacturing
- Power equipment / nuclear supply chain

### Secure Sufficient AI Capability

- Energy & data centers
- Models & data, compute, cloud
- Security and safety capabilities

### Build Next Strengths

- Physical AI — manufacturing, robotics
- Bio AI — healthcare, CDMO, etc.
- AI semiconductors - accelerators, edge AI, advanced packaging
- With experts in global AI hubs



# V. How Korea might Position Itself in the AI Era

## 2 ALLY – Build with the Best




### Critical Partners, Critical Technologies

- Co-build frontier models, chips, cloud, and safety with trusted partners
- Plug Korean firms into global AI stacks and technical standards
- Gain access to talent, compute, markets, and frontier know-how

## 3 ADOPT – Use the Best Technology Faster



### Rapid Diffusion of Frontier Technologies

- Move from pilots to economy-wide adoption: SMEs, factories, hospitals, schools, government
- Create sandboxes, procurement channels, and testbeds for safe deployment
- Build broad AI literacy to unlock innovation across society and the economy

☞ **Silicon Valley's Korean tech community can be a bridge between frontier AI and Korea's industries.**

# V. How Korea might Position Itself in the AI Era

## 4 RULE – Help Shape the Rules of the AI Era

- If AI is becoming infrastructure, security, labor, data, and platform policy at once,
  Korea should not only follow rules — it should help shape them.

### Where Korea Can Help Shape Rules

- Cyber and bio security
- AI safety, evaluation, and agent liability
- Data, copyright, and platform governance
- AI compute, and semiconductor interoperability standards
- Robotics, autonomous systems, and human-machine safety




**Standards do not simply follow markets,
Sometimes standards create markets.**

☞ **OWN** what matters. **ALLY** where others are stronger. **ADOPT** faster. **RULE** while the rules are still being written.

# A useful analogy: The Age of Exploration

## 1. Many Supply Chains

| Age of Exploration | AI Era |
|---|---|
| Ships | **Models, chips, and compute** |
| Sea routes | **Cloud, platforms, and distribution** |
| Ports & trade networks | **Ecosystems, alliances, and adoption** |
| Rules of the sea | **Governance, standards, and sovereignty** |

## 2. Where does Value go?

**Age of Exploration**

Pepper → Silver → Cotton textiles & tea → Sugar & Production

**AI Era**

Chips & compute → Cloud, models, apps → Physical AI, Bio AI, enterprise?

**Early winners may not be the final winners.**

## 3. Leaders can Change

Portugal / Spain → Dutch → Britain

**AI leadership may also shift.**

## 4. Side Effects and Risks

| Exploration | AI |
|---|---|
| • Violence<br>• Colonial extraction<br>• Slavery<br>• Disease spread | • Labor disruption<br>• Power concentration<br>• Security & misuse<br>• Autonomy risks |

☞ **Like the Age of Exploration, AI may not just create new products — it may reshape trade, power, and world order. We do not know all the winners yet, but we should move early, manage risks, and position wisely.**